

English
Medium



EDU TERIA

72nd BPSC

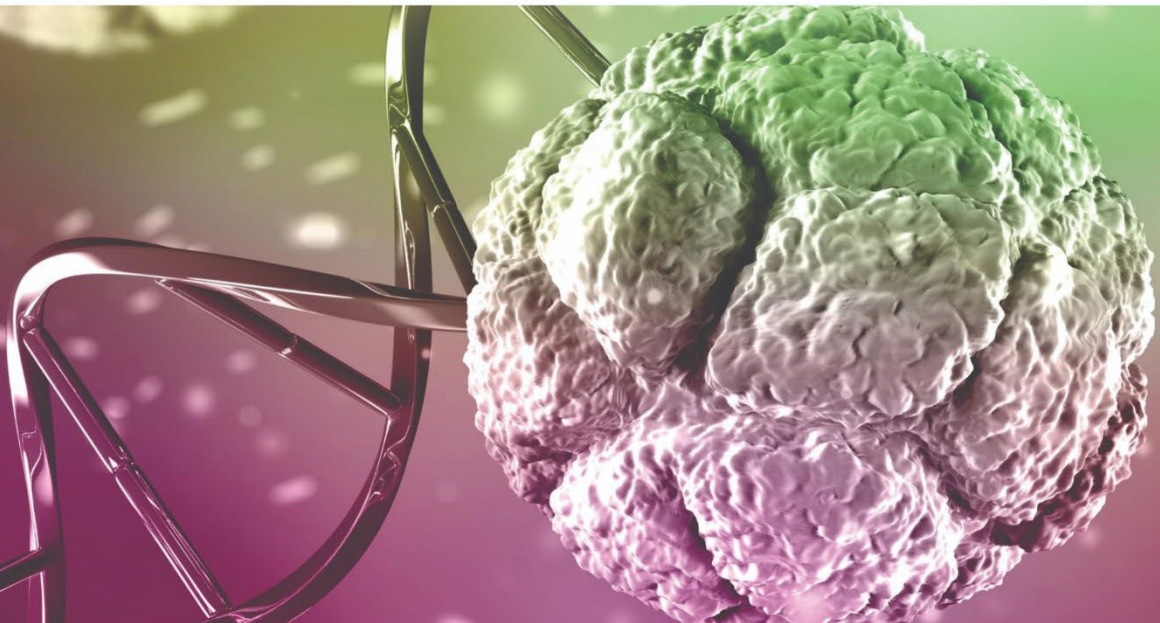
Prelims Online Batch

BIOLOGY

LECTURE

14

Digestive System, Part - 02



Digestive System, Part - 02

Digestive Glands

- Exocrine in nature
- Responsible for secretion of Enzymes.
- Made up epithelial tissue.

(i) Salivary Gland—(Saliva)

(a) **Parotid Gland (1 pair)** — Largest salivary gland secretes saliva with enzyme known as “Ptyalin” present below both ears.

(b) Submaxillary or Submandibular (1 pair)—

Present at lower Jaw Secretes Saliva.

(c) Sublingual- Present below the tongue secretes saliva.

(2) Gastric gland—Secretion is known as gastric juice.

It contains, mucus, HCl, pepsinogen and prorennin.

Enzyme	Site of Action	Substrate	Products of Action
Salivary Juice (Salivary Gland)			
Salivary amylase or Ptyalin	Buccal cavity	Starch	Disaccharides (few)
Gastric Juice (Stomach)			
Pepsin	Stomach	Proteins	Large peptides
Pancreatic Juice (Pancreas)			
Pancreatic α-amylase	Small intestine	Starch	Disaccharides
Trypsin	Small intestine	Proteins	Large peptides
Chymotrypsin	Small intestine	Proteins	Large peptides
Carboxypeptidases	Small intestine	Large peptides	Amino-acid
Lipase	Small intestine	Triglycerides	Monoglycerides fatty acids,
Nucleases	Small intestine	Nucleic acids	Nucleotides
Intestinal Juice (Small Intestine)			
Enteropeptidase or enterokinase	Small intestine	Trypsinogen	Trypsin
Aminopeptidase	Small intestine	Large peptides	Amino-acid
Peptidase	Small intestine	Oligopeptides	Amino acids
Disaccharidases	Small intestine	Disaccharides	Monosaccharides
Nucleotidase	Small intestine	Nucleotides	Nucleosides phosphoric acid
Nucleosidases	Small intestine	Nucleosides	Sugars, purines pyrimidines
Lipase	Small intestine	Triglycerides	Monoglycerides, glycerol, fatty acids

(3) **Liver**—Largest gland, made up of Hepatic lobules (structural & functional unit of liver)

Weight – 1.5 – 1.7 kg

Secretion is known as bile juice.

pH=8 (Basic in nature)

Bile Juice store in gall bladder.

Bile Juice		
Bile pigment	Bile Salt	
● Provides colour to bile juice.	Inorganic Salt	Organic Salt
● Light greenish yellow colour.	NaH ₂ PO ₄	Sodium Glycolate
● Colour is due to bile pigment known as bilirubin and Biliverdin	Na ₂ CO ₃	Sodium taurocholate
	NaCl	

(4) **Pancreas**— Mixed gland: Made up to exocrine as well as endocrine cells.

It can secrete enzyme as well as Hormone also.

99% Cells are exocrine in nature.

Secretion of pancreatic gland known as pancreatic Juice. pH- 7.5 (Basic)

Physiology of Digestion

(i) **Oral Cavity:** Entry of food into oral cavity is known as ingestion.

Food mixes with saliva and gets fragmented into smaller pieces with the help of set of teeth fragmentation of food by mastication of teeth.

Food + Saliva— Bolus.

Saliva—pH-6.8 99% water, 1% enzyme-Electrolytes salts.

(2) **Pharynx**— This air is the common pathway of food and no digestion takes place.

(3) **Oesophagus**— Tube like muscularis structure connects oral cavity to stomach Bolus moves in esophagus by peristalsis movement and reaches into the stomach. No digestion takes place.

(4) **Stomach**— Bolus+Gastric juice—Chyme (Acidic in nature)

Chyme—Protein Pepsin Peptones + proteoses + peptide.

Infant milk Rennin Curd (Insoluble Paracaseinate)

(5) **Small Intestine**—



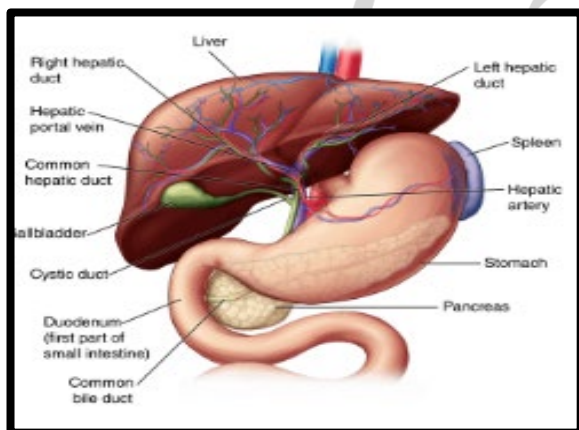
(i) **Duodenum**–(Mori mum digestion) wall of duodenum secretes secretin hormone, responsible for stimulation of bile Juice and pancreatic Juice accretion.

Digestive Glands

- The digestive glands associated with the alimentary canal include the salivary glands, the liver and the pancreas.
- Salivary glands Saliva is mainly produced by three pairs of salivary glands, the parotids (cheek), the sub-maxillary (lower jaw) and the sub-lingula's (below the tongue).
- These glands situated just outside the buccal cavity secrete salivary juice into the buccal cavity.
- The saliva breaks down the starch into sugar.

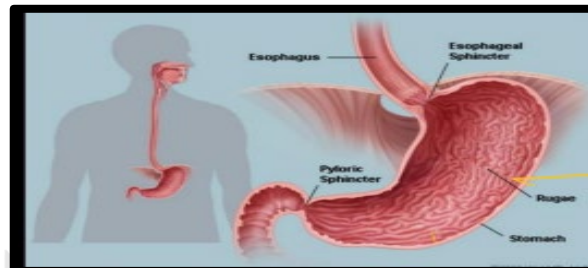
Liver

- The liver is a reddish-brown gland situated in the upper part of the abdomen on the right side.
- It is the largest gland in the body.
- It secretes bile juice that is stored in a sac called the gall bladder.
- The bile plays an important role in the digestion of fats.
- It has two lobes.
- The hepatic lobules are the structural and functional units of liver containing hepatic cells.
- The bile secreted by the hepatic cells passes through the hepatic ducts and is stored and concentrated in a thin muscular sac called the gall bladder.
- The duct of gall bladder (cystic duct) along with the hepatic duct from the liver, forms the common bile duct.



Stomach

The stomach is a muscular, J-shaped organ in the upper left abdomen that acts as a vital part of the digestive system, storing food and breaking it down through muscular contractions ([peristalsis](#)) and secreting enzymes and stomach acid.



Digestion in Stomach

Gastric Glands				
(Mucus Cell)	Neck	(Oxyntic Cell)	(Peptic or Chief Cell)	

Gastric Gland

- Gastric Juice
- pH = 1.5– 2.5
- (i) Mucus neck Cell–Secret mucus which present direct contact of HCl from inner wall of stomach.
- (ii) Oxyntic Cell/Parietal Cell–Secrets hydrochloric Acid.

Functioning of HCL

- (i) Kills microbes present in food particles
- (ii) Responsible for the activation of digestive enzymes (Primary Function)
Pepsinogen (Inactive) HCL pepsin
Prorenin (inactive) HCL Rennin
- (iii) **Peptic Cell/Chief Cell**– Responsible for the secretion of pepsinogen and prorenin.

Pepsinogen/Prorenin– Protein Digestive Enzymes

Duct system of liver, Pancreas & Gall bladder

